EBU6304 Software Engineering Group Project

2025-26

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First report

1. Requirements

1.1 Introduction and Project Scope

The BUPT International School Teaching Assistant Recruitment System is an Agile-based software solution that digitizes the current manual.Its scope covers Teaching Assistants, Module Organisers and Administrators , with technical constraints requiring it to be a standalone Java application using only plain text, JSON and CSV files for data persistence, without external databases or complex frameworks to focus on fundamental software engineering principles and legacy data interoperability.

1.2 Fact-Finding Techniques

To move beyond high-level abstractions and identify concrete system requirements, our team utilized the following fact-finding techniques:

1.2.1 Stakeholder Interview

We conducted structured interviews with potential Module Organisers (MOs, academic staff) to identify current bottlenecks in TA recruitment.

- **Fact-Finding Method**

To collect authentic requirements for the BUPT International School TA Recruitment System, we used one-to-one interviews with an MO directly responsible for TA recruitment, along with a targeted questionnaire. This approach captured real workflow pain points, functional demands, and operational habits that written documents cannot provide.

- **Interview Purpose**

The interview aimed to: clarify MO-led TA recruitment workflows; identify core needs in position publishing, applicant screening, and result management; define mandatory functions, constraints, and usability expectations for MOs; guide user story creation and prioritization for the product backlog.

- **Interview Process & Participants**

We interviewed one MO directly responsible for TA recruitment in a one-to-one format, following the questionnaire structure: recruitment process → position publishing → applicant screening → result management. The interviewee shared details based on daily operations.

- **Key Findings (MO-Focused)**
  - **Core Workflow**: MO submits TA position demand → admin approves → MO publishes positions → students apply → MO screens/interviews → MO confirms results → admin archives records.
  - **Position Publishing**: Mandatory information includes working hours, recruit number, deadline, location, and job requirements. The system must mark academic/major/skill requirements for different TA types and support position quantity limits (per school quota).
  - **Applicant Screening**: Core focus: major, experience, and grades (Course TA); time availability and responsibility (Invigilation TA). The system needs to record screening status and support sorting/recommending applicants by matching degree.

- **Impact on Requirements & Backlog**

Interview results defined MO-related user stories and priorities, confirming core functions (position publishing/modification, applicant screening, result confirmation synced with admin) to meet MOs' actual needs.

1.2.2 Questionnaire

- **Overview of the Questionnaire**

As part of the fact-finding process, a questionnaire was conducted to gather requirements from student users of the Teaching Assistant (TA) recruitment system, with 124 valid responses collected.

The questionnaire aims to:

a. Understand students' expectations and behaviours when applying for TA positions

b. Identify key system features

c. Support user story development and feature prioritisation with quantitative data

It mainly consists of closed-ended questions, enabling efficient analysis of user preferences. The questions cover job search behaviour, application process expectations, post-acceptance needs, and interest in advanced features.

- **Analysis of Questionnaire Results**

The questionnaire results are analysed based on the distribution of responses and the proportion of users selecting each option.

As shown in Figure, filtering is the most preferred method for searching TA opportunities (40%), followed by browsing (33%), while keyword search (15%) and recommendations (10%) are less popular. This suggests that students prefer searching based on specific criteria rather than general methods. Therefore, the system should support efficient filtering, such as by module, schedule, and requirements, to improve usability and search efficiency.

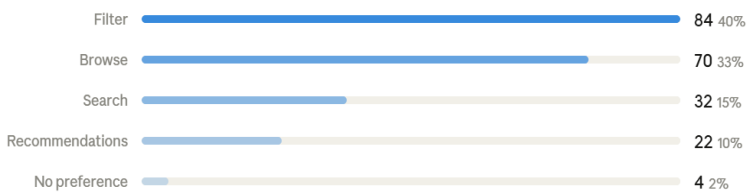

Figure 1. Preferred Methods of Students for Searching TA Positions

As shown in Figure, most students expressed interest in AI-assisted features. Job recommendations based on personal skills (103 responses) were the most preferred, followed by skill gap analysis (94 responses), while only a few (11) considered them unnecessary. This indicates that students are

generally open to intelligent support, especially for identifying suitable opportunities. However, despite this demand, these features are considered enhancements rather than core functions, and are better implemented in later iterations as “Could-have” features after the core workflow is established.

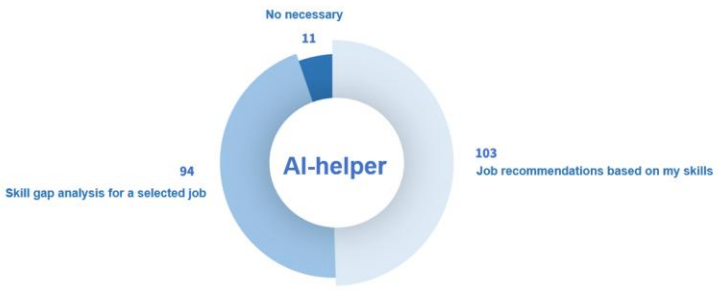

Figure 2. Students' Preferences for AI-Assisted Features

As shown in Figure X, checking application status (110 responses) is the most common action after submission, indicating a strong need for progress tracking. Viewing accepted jobs (75) and withdrawing applications (67) are also notable, showing the importance of application management. Other actions, such as editing or contacting organisers, are less frequently selected.

These findings suggest that application tracking and management should be prioritised as core (Must-have) features to ensure transparency and user control.

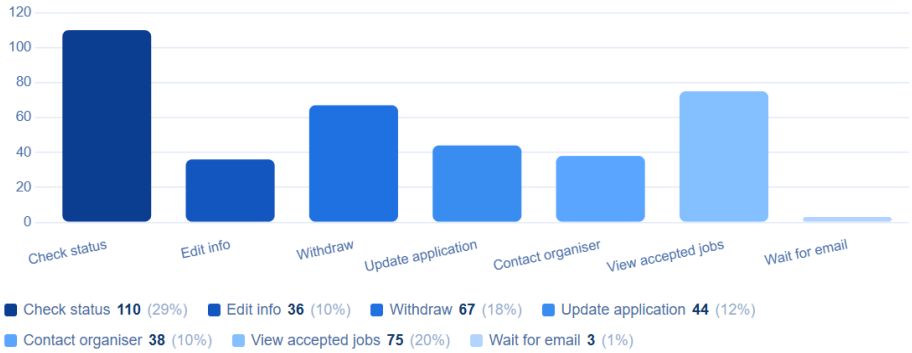

Figure 3.Students' Preferred Actions about the TA system

Questionnaire results highlight that students prioritise a clear and efficient application process, including job search and application tracking. They also want control over applications, such as withdrawing or viewing accepted positions. While AI-assisted features like job recommendations are appreciated, they are secondary to the core workflow. Development should first focus on a functional recruitment system, then gradually add enhancements based on user demand.

1.3 Prioritization Methods

To prioritise the product backlog for the TA Recruitment System, our team adopted a MoSCoW-based prioritisation approach. In Agile development, backlog items should be prioritised according to business value, expected return, and project risk, so that the most essential functionality is delivered first. The MoSCoW method is particularly suitable for this project because it helps distinguish between features that are indispensable for the system to operate and those that provide additional convenience or enhancement only.

In this project, each user story was assigned one of four priority levels:

- **Must Have = 1 point**

These are core features without which the system would not function properly. For example, login, job posting, applying for jobs, and checking application status are fundamental to the TA recruitment workflow.

- **Should Have = 3 points**

These features are important and valuable, but the system can still operate without them in an early release. Examples include filtering jobs, resetting passwords, or viewing assigned TA jobs.

- **Could Have = 5 points**

These stories improve usability or efficiency but are not essential to the system's basic operation. Examples include intelligent recommendations or workload warnings.

- **Want to Have = 7 points**

Interview results defined MO-related user stories and priorities, confirming core functions (position publishing/modification, applicant screening, result confirmation synced with admin) to meet MOs' actual needs.

These low-priority enhancements have limited business value and are deferred unless time permits. This prioritisation focused the team on features supporting the main recruitment workflow for TAs, Module Organisers, and Admins, reducing delivery risk by implementing essential stories first and postponing advanced/optional ones.Additionally, the scoring mechanism made prioritisation more actionable than labels alone: converting priority levels to numeric values combined priority with development effort, enabling transparent, consistent sprint allocation.

1.4 Estimation Method

For effort estimation, the team used story points based on the Fibonacci sequence: **1, 2, 3, 5,** and **8.** This approach is widely used in Agile projects because story points measure the relative effort, complexity, and uncertainty of implementing a user story, rather than exact hours. Larger tasks usually involve disproportionately greater uncertainty and integration risk, so a non-linear scale is more realistic than a linear one.

The meaning of the story points in this project is as follows:

- **1 point** – very small and straightforward work, with low uncertainty
- **2 points** – small feature with limited logic or UI work
- **3 points** – moderate feature involving several interactions or validations
- **5 points** – relatively complex feature requiring multiple classes, interfaces, or data operations
- **8 points** – highly complex feature with greater uncertainty, broader impact, or more difficult testing needs

The team estimated each story collaboratively by considering several factors:

- **Functional complexity** – how much logic the story requires
- **Technical difficulty** – whether the implementation involves unfamiliar code, multiple modules, or more complicated validation
- **Dependencies** – whether the story depends on other features being completed first
- **Testing effort** – whether the feature requires more extensive unit, integration, or regression testing

- **Risk and uncertainty** – whether there are unknowns in implementation or behaviour

This estimation approach is consistent with Agile principles, where user stories are selected for a release based on both priority and schedule estimates, and then broken down into implementation tasks by the development team.

To make prioritisation and sprint allocation objective, we combined priority and estimation into a single value:

- **Total Score = Priority Score + Estimation Score**

A lower total score indicates that a story is both more important and easier to deliver early, so it should be scheduled earlier in the project. This method helped the team avoid two common problems:first, implementing low-value features too early; second, overloading early sprints with very complex stories that might delay delivery.

1.5 Iteration Planning

The project adopted an Agile iterative development approach, where development is divided into short (typically two-week) iterations: the team selects user stories, breaks them into tasks, develops, integrates, tests, and evaluates before proceeding to the next iteration.

The project backlog was divided into four sprints, allocated by each story's total score:

- Total Score ≤ 4 → Sprint 1
- Total Score ≤ 6 → Sprint 2
- Total Score ≤ 8 → Sprint 3
- Total Score > 8 → Sprint 4

This ensured high-priority, low-effort stories were delivered first, while lower-value or more complex ones were deferred.

- **Sprint 1**

Focused on the minimum viable product (MVP) and core TA recruitment workflow (stories with total score ≤ 4). It established the system foundation (core account access, key interactions for Applicants and Module Organisers), enabling basic recruitment actions and demonstrating core business value.

- **Sprint 2**

Included stories with total scores 5–6 (important but slightly more complex/less urgent than Sprint 1). It extended the system with features like application submission, hiring decisions, password support, and workload management, building on Sprint 1's foundation.

- **Sprint 3**

Covered stories with total scores 7–8 (supporting/optimisation features). Its goal was to improve efficiency, visibility, and user experience, including notifications, data export, historical records, filtering, and non-functional enhancements (e.g., performance).

- **Sprint 4**

Contained stories with total score > 8 (most complex/lowest-priority, e.g., advanced intelligent features). These were deferred to reduce risk of disrupting core system delivery.

This strategy aligns with Agile principles of incremental delivery and early high-value delivery, ensuring manageable workflows, step-by-step integration/testing, and controlled technical risk.

2. Supporting material

2.1 Interview Script (Target: Module Organisers)

2.1.1 Interview Questions (From Students to Teachers in Charge of School TA Recruitment)

- **Basic Recruitment Process and Post Publishing Entity**

  Q1. Hello teacher, we'd like to ask about the overall process of the school's TA recruitment each semester. What are the core key steps?

  Q2. What main categories are the school's TA positions divided into? What are the core differences in recruitment requirements for different positions?

- **Post Publishing and Information**

  Q1. What core mandatory information do teachers need to fill in when publishing TA positions?

  Q2. Do the academic qualification, major and skill requirements for different TA positions need to be clearly marked and support screening in the system?

  Q3. Does the school set a limit on the number of TA positions? Does the system need to add a function for setting the number of positions?

- **Applicant Screening and Employment**

  Q1. What core information do you focus on most when screening TA applicants?

  Q2. What is the screening process? Does the system need to record the screening status of applications (e.g., pending review, employed, etc.)?

  Q3. When multiple applicants apply for the same position, does the system need to sort and recommend them by matching degree?

- **Administrator and Workload Management**

  Q1. Does the school have special teachers responsible for TA workload statistics? What are the core statistical dimensions?

  Q2. Is there an upper limit on TA workload? Does the system need to add a workload early warning or restriction function?

  Q3. Do administrators need to view all position-related data and export it for filing?

- **TA Applicant Operations and System Experience**

  Q1. Besides creating profiles and applying for positions, what other basic operation functions do TA applicants need?

  Q2. Is it sufficient for the application status to be displayed only in the system, or is an additional message reminder required?

  Q3. Does the system interface tend to be simple and easy to use or fully functional? What are the requirements for data backup?

  Q4. Besides basic functions, what other additional suggestions do you have for the system?

2.1.2 Simulated Interview Content (Conversation Between Student Team and Teacher in Charge of School TA Recruitment)

Student team representative: Hello teacher, we are a student team majoring in Software Engineering at the school. We are currently developing the school's TA recruitment system. Today we want to consult you about the actual recruitment needs, so that we can make the system more suitable for the school's usage scenarios. Sorry to take up a little of your time.

School teacher: No problem, feel free to ask. I will tell you all the actual situations in recruitment, and I also hope the system you develop can really simplify our current recruitment process.

Student team representative: Thank you so much, teacher! First of all, we'd like to ask about the overall process of the school's TA recruitment each semester. What are the core key steps?

School teacher: Our process is actually very clear. The first step is that the Module Organizer (MO) of each course submits position requirements to me; after I approve them, the MO publishes the positions. The second step is that students submit applications, and the MO collects application materials. The third step is that the MO screens resumes and may conduct a simple online interview for qualified students. The fourth step is that the MO finalizes the employment list, and I keep the unified records for filing. Finally, the MO notifies the employed students and arranges their onboarding. At present, we all use Excel and forms; the steps are simple but the collation work is extremely troublesome.

Student team representative: We understand your pain points, teacher. We will fully integrate this process into the system later. Then what main categories are the school's TA positions divided into? What are the core differences in recruitment requirements for different positions?

School teacher: They are mainly divided into two categories. The first is **Course Teaching Assistant TA**, who helps the MO with homework correction, student tutoring, and course material organization. This type requires a high degree of major matching, preferably students from the same or related majors; some courses also require good English skills to read foreign language materials. The second is **Invigilation TA**, who mainly cooperates with the MO to complete course invigilation work. This type requires carefulness, a strong sense of responsibility and ample time, with no special major requirements. The MO is responsible for the recruitment and management of both types of teaching assistants.

Student team representative: Alright, we will add an option for position types in the system and set personalized qualification requirement fields for different positions. In addition, we'd like to ask, is the MO the only entity to publish TA positions? Do the operation processes need to be distinguished when MOs publish different types of positions?

School teacher: Yes, all positions are published by the course MOs. Both course teaching assistant and invigilation TA positions are published by the corresponding responsible MOs, as they are all core work related to teaching assistants. The operation processes do not need to be distinguished and should be kept unified. This way, it will be more convenient for MOs to use, and they don't need to adapt to different operation methods additionally.

Student team representative: Understood! We will develop the system with the MO as the sole publishing entity throughout the process, and keep the publishing process of all position types unified to fit the MOs' usage habits and also reduce development workload. Then what core mandatory information do teachers need to fill in when publishing positions?

School teacher: The mandatory items include **working hours** (e.g., hours per week), **number of recruits**, **application deadline for the position**, and **work location** (online/offline). The salary follows the school's unified standard and does not need to be filled in separately; it can be displayed fixed in the system. In addition, the **qualification requirements** must be clear, such as major, skills, and grade.

Student team representative: Got it, we will set these as mandatory fields to avoid missing information. Then what core information do teachers and MOs focus on most when screening applicants?

School teacher: When screening course teaching assistant TAs, MOs focus most on **major matching degree** and **previous TA experience**, and relevant course grades are a plus. For invigilation TAs, the key factors are **ample time** and **personal basic qualities**, after all, they need to cooperate with the MO to complete course invigilation work. When I coordinate the overall recruitment, I will pay attention to the **number of positions each TA applies for** to prevent someone from applying for too many and being unable to manage the workload.

Student team representative: Then we will highlight these core information in the applicant list of the system for the convenience of MOs' screening. Another question, does the system need to record the application status after screening? For example, pending review, passed the initial screening, etc.

School teacher: Of course it's needed. At present, we manually mark this in Excel, which is extremely easy to get messy. If the system can update the status automatically, applicants can check it by themselves, and we won't have to reply to students' inquiries repeatedly, which will save a lot of time and effort.

Student team representative: No problem, we will develop this status inquiry function as a core function. Next, we'd like to ask about workload management. Does the school have special teachers responsible for TA workload statistics? What are the core statistical dimensions?

School teacher: I am responsible for the statistics, with the core dimensions being **cumulative working hours** and **number of positions participated in**. The school has a regulation that an undergraduate student's TA working hours per semester shall not exceed 20 hours, and a postgraduate student's shall not exceed 30 hours, to avoid affecting students' studies.

Student team representative: That's crucial. We will add a workload statistics function in the system and set an **upper limit early warning**. If a TA's applied workload is approaching the upper limit, the system will remind the administrator and also restrict them from applying for more positions, so that teachers don't have to calculate manually.

School teacher: This function is particularly practical; it would be great if you can develop it! In addition, I need to be able to view the overall situation of all positions, such as which positions have been published, which have completed employment, and which are still open for applications. It would be better to export the data into an Excel spreadsheet for my filing work.

Student team representative: Rest assured, teacher. The administrator terminal will have a data summary and export function, supporting CSV/Excel formats to meet the filing requirements. Then for TA applicants, what other basic operation functions do you think they need besides the basic profile creation and position application?

School teacher: At the very least, they should be able to **modify personal information** and **withdraw applications**. For example, if a student fills in the wrong major or has no time to apply temporarily, they can operate it by themselves without coming to us again. In addition, **viewing historical application records** is needed, so that students can know which positions they have applied for before and the results.

Student team representative: Alright, we will add all these functions. Then is it sufficient for the application status to be displayed only in the system, or is an additional message reminder required?

School teacher: Displaying it in the system is enough; students can check it as long as they log in. Message reminders are easy to be ignored instead, so there's no need to make it too complicated. In addition, MOs can **voluntarily add notes on the reasons for non-employment** for unsuccessful applicants, and it is not mandatory. Some MOs may not have time to write, so it's better to keep it flexible.

Student team representative: Understood, we will add an optional note column for the reasons for non-employment to balance flexibility. Then when using office systems daily, do you prefer the interface to be simple and easy to use or fully functional? What are the requirements for data backup?

School teacher: Definitely a **simple operation interface**. We teachers have a lot of work to do every day, and it's troublesome to learn an overly complicated system. Core functions should be placed in prominent positions. For example, after MOs log in, they can directly see "Publish Position" and "Screen Applicants", and the fewer operation steps, the better.

Student team representative: We will pay close attention to this point, and the system design will adhere to the principle of being simple and easy to use. What about the requirements for data backup at the school?

School teacher: Data must be backed up regularly, such as **automatic weekly backup**. The backup format can be JSON or Excel, and the data should be stored on the school's computers to avoid data loss due to system problems. It would be fine to add a one-click backup function in your system.

Student team representative: Alright, we will add both manual and automatic backup functions to meet data security requirements. Finally, we'd like to ask if you have any additional expectations or suggestions for the system besides what we discussed today?

School teacher: Not many additional suggestions. I just hope the system can be **compatible with the school's computer systems** and can be opened on both Windows and Mac. In addition, there should be simple prompt words during operation, such as a prompt of "Please fill in your real major" when entering information, to prevent students from filling in wrong information. It would be even better if a simple **skill matching function** can be added in the later stage. For example, if a course teaching assistant position published by an MO requires "Python skills", the system can automatically rank applicants who master Python at the top, which can save MOs a lot of screening time.

Student team representative: We will adopt all these suggestions, teacher. The system will be made compatible with different systems and add operation prompt words. We will also develop the simple AI skill matching as an extended function, prioritizing the stability and ease of use of basic functions. Thank you so much for your time today, teacher. Your answers have made our development direction of the system clearer. After the system development is completed, we will invite you to test it and see what needs to be improved.

School teacher: No problem. You can come to me again if you have any questions in the follow-up. I hope your system can be completed soon to solve our current recruitment pain points.

Student team representative: We will definitely work hard. Thank you again for your cooperation and guidance, and wish you a smooth work!

School teacher: You're welcome, keep it up!

2.2 Questionnaire Script (Target: Students)

2.2.1 Questionnaire content

Q1. Which of the following best describes your current year of study?

- A. Year 2
- B. Year 3
- C. Year 4
- D. Master's Year 1
- E. Master's Year 2
- F. Master's Year 3
- G. other

Q2. Have you ever applied for a Teaching Assistant (TA) position before?

- A. Yes
- B. No

Q3. How interested are you in applying for TA positions in the future?

- A. Very interested
- B. Interested
- C. Neutral
- D. Not interested

Q4. Which information should be shown on the job list page?

- A. Module name
- B. Required skills
- C. Workload
- D. Semester
- E. Application deadline
- G. other

Q5. How would you prefer to look for TA opportunities in the system?

- A. Browse all available jobs
- B. Search by keywords
- C. Filter by conditions
- D. Receive recommendations

- E. other

Q6. Which filtering options would be most useful when searching for TA positions? (Q5.C-->Q6)

- A. Module name
- B. Required skills
- C. Workload
- D. Semester
- E. Schedule / time
- G. other

Q7. How would you prefer to apply for a TA position?

- A. Directly through the system
- B. Through email after viewing the job
- C. Through an external form link
- D. No preference

Q8. After submitting an application, what would you most want to do in the system?

- A. Check application status
- B. Edit application information
- C. Withdraw application
- D. Replace or update application
- E. Contact the module organiser
- F. View my accepted TA jobs in one place
- G. other

Q9. If you are accepted for a TA position, what information should the system show you?

- A. Assigned TA jobs
- B. Module name
- C. Workload
- D. Work schedule
- E. Responsibilities
- G. other

Q10. Which intelligent feature would be more valuable to you?

- A. Job recommendations based on my skills
- B. Skill gap analysis for a selected job
- C. No necessary

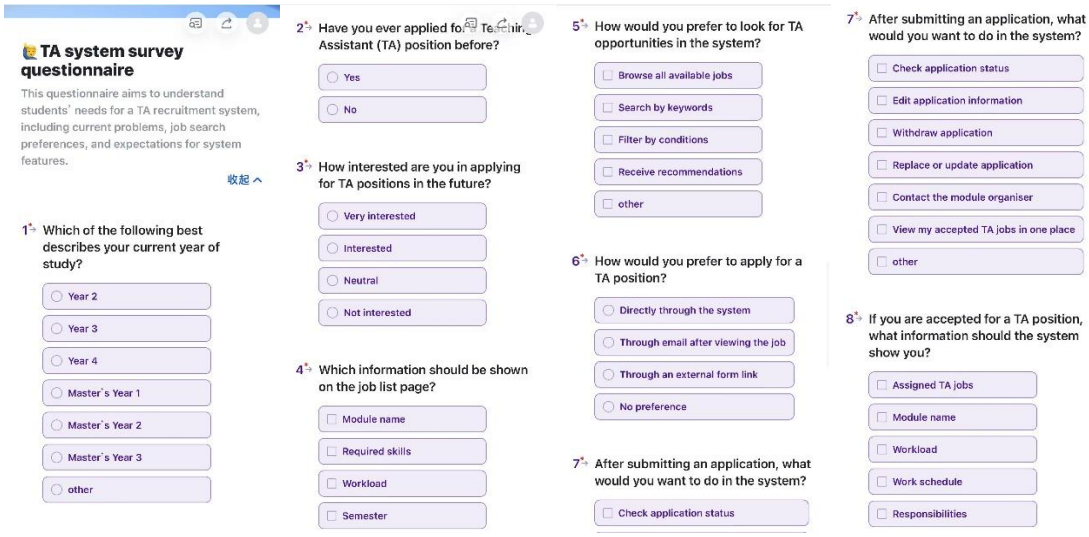

Figure 4. Feishu Questionnaire Chart

2.2.2 Questionnaire results

Q1. Which of the following best describes your current year of study?

Option	Count	Percentage
Year 2	18	14.5%
Year 3	28	22.6%
Year 4	40	32.3%
Master's Year 1	23	18.5%
Master's Year 2	11	8.9%
Master's Year 3	2	1.6%
PhD Year 1	2	1.6%
Total	**124**	**100%**

Table 1. Q1 result

Q2. Have you ever applied for a Teaching Assistant (TA) position before?

Option	Count	Percentage
Yes	44	35.5%
No	80	64.5%
Total	**124**	**100%**

Table 2. Q2 result

Q3. How interested are you in applying for TA positions in the future?

Option	Count	Percentage

Very interested	36	29.0%
Interested	54	43.5%
Neutral	22	17.7%
Not interested	12	9.7%
Total	**124**	**100%**

Table 3. Q3 result

Q4. Which information should be shown on the job list page?

Option	Count
Application deadline	115
Module name	110
Workload	102
Required skills	96
Semester	78
Number of positions	36
Location	12

Table 4. Q4 result

Q5. How would you prefer to look for TA opportunities in the system?

Option	Count	Percentage
Filter	84	67.7%
Browse	70	56.5%
Search	32	25.8%
Recommendations	22	17.7%
No preference	4	3.2%

Table 5. Q5 result

Q6. Which filtering options would be most useful when searching for TA positions? (Q5.C-->Q6)

Option	Count
Module name	38
Schedule / time	36
Required skills	34
Workload	30
Semester	18

Location	7

Table 6. Q6 result

Q7. How would you prefer to apply for a TA position?

Option	Count	Percentage
System	88	71.0%
Email	18	14.5%
External form	12	9.7%
No preference	6	4.8%

Table 7. Q7 result

Q8. After submitting an application, what would you most want to do in the system?

Option	Count
Check status	110
View accepted jobs	75
Withdraw	67
Update application	44
Contact organiser	38
Edit info	36
Wait for email notification	3

Table 8. Q8 result

Q9. If you are accepted for a TA position, what information should the system show you?

Option	Count
Schedule	104
Assigned jobs	98
Workload	92
Responsibilities	88
Module name	84
Deadline reminders of rejection	33
Contact details of lecturer	4

Table 9. Q9 result

Q10. Which intelligent feature would be more valuable to you?

Option	Count	Percentage
Job recommendation	103	83.1%
Skill gap analysis	94	75.8%
Not necessary	11	8.9%

Table 10. Q10 result